

End Effector Command set (optional)

CAUTION

The following interfaces are **only applicable** to the INYU EG2-4C2 gripper.

Gripper control

The RealMan robotic arm is equipped with the INSPIRE-ROBOTS EG2-4C2 gripper. To facilitate user operation of the gripper, the robotic arm controller provides an open gripper control protocol (the gripper control protocol is mutually exclusive with the end modbus functionality).

Set the gripper stroke `set_gripper_route`

Set the gripper stroke, which refers to the maximum and minimum opening values of the gripper. Once set successfully, the values are saved automatically and will not be lost when the gripper is powered off.

- **Input parameter**

Parameter	Type	Description
<code>set_gripper_route</code>	<code>string</code>	Set the gripper stroke.
<code>min</code>	<code>int</code>	Minimum opening value of the gripper, range: 0–1,000, without a unit of measurement.
<code>max</code>	<code>int</code>	Maximum opening value of the gripper, range: 0–1,000, without a unit of measurement.

- Code demo

Input

```
{ "command": "set_gripper_route", "min": 70, "max": 500 }
```

json

Output

Setting succeeded:

```
{
  "command": "set_gripper_route",
  "state": true
}
```

json

Setting failed:

```
{
  "command": "set_gripper_route",
  "state": false
}
```

json

 Set the gripper release `set_gripper_release`

- Input parameter

Parameter	Type	Description
<code>set_gripper_release</code>	<code>string</code>	Set the gripper release.
<code>speed</code>	<code>int</code>	Release speed of the gripper, range: 1–1,000, without a unit of measurement.
<code>block</code>	<code>int</code>	true: blocking mode, false: non-blocking mode.

- Output parameter

Parameter	Type	Description
<code>state</code>	<code>bool</code>	<code>true</code> : setting succeeded, <code>false</code> : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

- **Code demo**

Input

Set the gripper release.

```
{"command":"set_gripper_release","speed":500,"block":true}
```

json

Output

Gripper release succeeded:

This command will be returned regardless of whether it is in blocking mode.

```
{
  "command": "set_gripper",
  "state": true
}
```

json

Gripper release failed:

This command will be returned regardless of whether it is in blocking mode.

```
{
  "command": "set_gripper",
  "state": false
}
```

json

This command is used to report the motion to a given position in blocking mode.

```
{"state":"current_trajectory_state","trajectory_state":true,"device":1,"trajectory_c
```

json

Set the force-controlled grasping of the gripper `set_gripper_pick`

Force-controlled grasping of the gripper: The gripper grasps with the set speed and force. When the gripping force exceeds the preset threshold, the gripper stops grasping.

• Input parameter

Parameter	Type	Description
set_gripper_pick	string	Set the force-controlled grasping of the gripper.
speed	int	Release speed of the gripper, range: 1–1,000, without a unit of measurement.
force	int	Force control threshold, range: 50–1,000, without a unit of measurement.
block	int	true: blocking mode, false: non-blocking mode.

• Output parameter

Parameter	Type	Description
state	bool	true : setting succeeded, false : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

• Code demo

Input Set the force-controlled grasping of the gripper.

```
 {"command": "set_gripper_pick", "speed": 500, "force": 200, "block": true} 
```

json

Output

Gripper release succeeded:
This command will be returned regardless of whether it is in blocking mode.

```
 {
  "command": "set_gripper",
  "state": true
} 
```

json

Gripper release failed:
This command will be returned regardless of whether it is in blocking mode.

json

```
{
  "command": "set_gripper",
  "state": false
}
```

This command is used to report the motion to a given position in blocking mode.

json

```
{
  "state": "current_trajectory_state",
  "trajectory_state": true,
  "device": 1,
  "trajectory_connect": 1
}
```

Set the continuous force-controlled grasping of the gripper



set_gripper_pick_on

Continuous force-controlled grasping of the gripper: The gripper grasps with the set speed and force. When the gripping force exceeds the preset threshold, the gripper stops grasping. When the gripping force is less than the preset threshold, the gripper grasps again until the gripping force exceeds the preset threshold.

• Input parameter

Parameter	Type	Description
set_gripper_pick_on	string	Set the continuous force-controlled grasping of the gripper.
speed	int	Release speed of the gripper, range: 1–1,000, without a unit of measurement.
force	int	Force control threshold, range: 50–1,000, without a unit of measurement.
block	int	true: blocking mode, false: non-blocking mode.

• Output parameter

Parameter	Type	Description
<code>state</code>	<code>bool</code>	<code>true</code> : setting succeeded, <code>false</code> : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

- **Code demo**

Input

Set the continuous force-controlled grasping of the gripper.

```
{"command": "set_gripper_pick_on", "speed": 500, "force": 200, "block": true}
```

json

Output

Gripper release succeeded:

This command will be returned regardless of whether it is in blocking mode.

```
{  
  "command": "set_gripper",  
  "state": false  
}
```

json

Gripper release failed:

This command will be returned regardless of whether it is in blocking mode.

```
{  
  "command": "set_gripper",  
  "state": false  
}
```

json

This command is used to report the motion to a given position in blocking mode.

```
{  
  "state": "current_trajectory_state",  
  "trajectory_state": true,  
  "device": 1,  
}
```

json

```
"trajectory_connect": 1
}
```

Set the gripper to a given position `set_gripper_position`

Gripper at a given position: When the current opening is smaller than the given opening, the gripper releases to a given position at the given speed. When the current opening is larger than the given opening, the gripper closes to a given position at the given speed and force. The gripper stops when the gripping force exceeds the preset threshold or when the given position is reached.

- Input parameter

Parameter	Type	Description
<code>set_gripper_position</code>	<code>string</code>	Set the gripper to a given position.
<code>position</code>	<code>int</code>	Opening position of the gripper, range: 1–1,000, without a unit of measurement.
<code>block</code>	<code>int</code>	true: blocking mode, false: non-blocking mode.

- Output parameter

Parameter	Type	Description
<code>state</code>	<code>bool</code>	<code>true</code> : setting succeeded, <code>false</code> : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

- Code demo

Input

Set the gripper to a given position.

```
{"command": "set_gripper_position", "position": 500, "block": true}
```

json

Output

Gripper release succeeded.

This command will be returned regardless of whether it is in blocking mode.

```
{
  "command": "set_gripper",
  "state": true
}
```

json

Gripper release failed.

This command will be returned regardless of whether it is in blocking mode.

```
{
  "command": "set_gripper",
  "state": false
}
```

json

This command is used to report the motion to a given position in blocking mode.

```
{
  "state": "current_trajectory_state",
  "trajectory_state": true,
  "device": 1,
  "trajectory_connect": 1
}
```

json

Get the gripper state `get_gripper_state`

- Input parameter

Parameter	Type	Description
<code>get_gripper_state</code>	<code>string</code>	Get the gripper state.

- Output parameter

Parameter	Type	Description
<code>enable</code>	<code>int</code>	Gripper enabling state, 0: disable, 1: enable.

Parameter	Type	Description
status	int	Gripper online state, 0: offline, 1: online.
error	int	Gripper error message. Indicated by the lower 8 bits. Bit5-7: reserved. Bit4: internal communication. Bit3: driver. Bit2: overcurrent. Bit1: over-temperature. Bit0: locked rotor.
mode	int	Current mode, 1: gripper fully open and idle, 2: gripper fully closed and idle, 3: gripper stopped and idle, 4: gripper in closing, 5: gripper in opening, 6: gripper stopped during closing due to force control.
current_force	int	Current force of the gripper, in g.
temperature	int	Current temperature, in °C.
actpos	int	Gripper opening.

• Code demo

Input

```
{ "command": "get_gripper_state" }
```

json

Output

```
{
  "command": "get_gripper_state",
  "enable": 1,
  "status": 1,
  "error": 0,
  "mode": 1,
  "current_force": 100,
  "temperature": 40,
  "actpos": 150
}
```

json

Five-fingered dexterous hand

The RealMan robotic arm is equipped with an INSPIRE-ROBOTS five-fingered dexterous hand, which can be configured by the protocol.

Set the gesture of the dexterous hand `set_hand_posture`

- Input parameter

Parameter	Type	Description
<code>set_hand_posture</code>	<code>string</code>	Set the gesture of the dexterous hand.
<code>posture_num</code>	<code>int</code>	Gesture sequence number pre-stored in the dexterous hand, range: 1-40.
<code>block</code>	<code>bool</code>	true: blocking mode, false: non-blocking mode.

- Output parameter

Parameter	Type	Description
<code>set_state</code>	<code>bool</code>	<code>true</code> : setting succeeded, <code>false</code> : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

- Code demo

Input

Set the gesture of the dexterous hand.

```
{ "command": "set_hand_posture", "posture_num": 1, "block": true }
```

json

Output Setting succeeded:

```
{  
  "command": "set_hand_posture",  

```

json

```
"set_state": true
}
```

Setting failed:

```
{
  "command": "set_hand_posture",
  "set_state": false
}
```

json

This command is used to report the motion to a given position in blocking mode.

```
{
  "state": "current_trajectory_state",
  "trajectory_state": true,
  "device": 2,
  "trajectory_connect": 1
}
```

json

Set the action sequence of the dexterous hand

set_hand_seq

• Input parameter

Parameter	Type	Description
set_hand_seq	string	Set the action sequence of the dexterous hand.
seq_num	int	Action sequence number pre-stored in the dexterous hand, range: 1–40.
block	bool	true: blocking mode, false: non-blocking mode.

• Output parameter

Parameter	Type	Description
set_state	bool	true : setting succeeded, false : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

- **Code demo**

Input

Set the action sequence of the dexterous hand.

```
{"command": "set_hand_seq", "seq_num": 1, "block": true}
```

json

Output Setting succeeded:

```
{
  "command": "set_hand_seq",
  "set_state": true
}
```

json

Setting failed:

```
{
  "command": "set_hand_seq",
  "set_state": false
}
```

json

This command is used to report the motion to a given position in blocking mode.

```
{
  "state": "current_trajectory_state",
  "trajectory_state": true,
  "device": 2,
  "trajectory_connect": 1
}
```

json

Set the angles of the dexterous hand **set_hand_angle**

Set the angles of the dexterous hand, 6 active degrees of freedom (DOF), DOF 1 (thumb flexion), DOF 2 (index finger), DOF 3 (middle finger), DOF 4 (ring finger), DOF 5 (little finger), DOF 6 (thumb rotation).

• Input parameter

Parameter	Type	Description
set_hand_angle	string	Set the finger angles.
hand_angle	int	Array of finger angles, range: 0–1,000. Additionally, -1 indicates that no action will be performed for this degree of freedom, and the current state will be maintained.
block	bool	true: blocking mode, false: non-blocking mode.

• Output parameter

Parameter	Type	Description
set_state	bool	true : setting succeeded, false : setting failed.

You can refer to [current_trajectory_state](#) for the output parameters of the motion completion return result.

• Code demo

Input

Set the angles of the dexterous hand.

```
{ "command": "set_hand_angle", "hand_angle": [-1, 100, 200, 300, 400, 500], "block": true }
```

json

Output Setting succeeded:

```
{
  "command": "set_hand_angle",
  "set_state": true
}
```

json

Setting failed:

```
{
  "command": "set_hand_angle",
  "set_state": false
}
```

json

```
"set_state": false
}
```

This command is used to report the motion to a given position in blocking mode.

```
{
  "state": "current_trajectory_state",
  "trajectory_state": true,
  "device": 2,
  "trajectory_connect": 1
}
```

json

Set the speed of the dexterous hand `set_hand_speed`

- Input parameter

Parameter	Type	Description
<code>set_hand_speed</code>	<code>string</code>	Set the finger speed.
<code>hand_speed</code>	<code>int</code>	Finger speed, range: 1–1,000.

- Code demo

Input

```
{"command": "set_hand_speed", "hand_speed": 500}
```

json

Output Setting succeeded

```
{
  "command": "set_hand_speed",
  "set_state": true
}
```

json

Setting failed

```
```json
{
```

```
"command": "set_hand_speed",
"set_state": false
}
```

 **Set the force threshold of the dexterous hand** `set_hand_force`

• **Input parameter**

Parameter	Type	Description
<code>set_hand_force</code>	<code>string</code>	Set the finger force threshold.
<code>hand_force</code>	<code>int</code>	Finger force, range: 1–1,000.

• **Code demo**

**Input**

```
{ "command": "set_hand_force", "hand_force": 500 }
```

json

**Output** Setting succeeded

```
{
 "command": "set_hand_force",
 "set_state": true
}
```

json

**Setting failed**

```
{
 "command": "set_hand_force",
 "set_state": false
}
```

json

Previous

Force Sensor Command Set (Optional)

Next

End-Effector Ecosystem Command Set  
(optional)